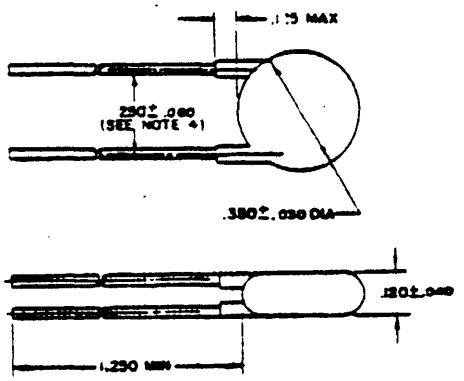
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING) (INSULATED), NON-ESTABLISHED RELIABILITY, STYLE CC51

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION. SEE MIL-C-20/44 FOR ITEMS PREVIOUSLY COVERED IN MIL-C-0020/17B(USAF).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-20.



INCHES	MM
.030	.76
.040	1.02
.060	1.52
.120	3.05
.125	3.18
.250	6.35
.350	8.89
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned wire of .025±.002(.64±.05 mm) diameter.
4. This dimension shall be measured at the point of emergence of the leads from the body of the capacitor.

FIGURE 1. Style CC51 capacitors (insulated).

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Lead type: Radial.
DC rated voltage: 500 volts.
Operating temperature range: -55° to +125°C.
Characteristic: See table I.

Capacitance: Within tolerance specified (see table I).

Quality factor (Q): In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.
Body insulation - Test IL

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Capacitance (pF)	Capacitance tolerance	Temperature coefficient shown as column head and tolerance envelope for temperature coefficient shown in column							
			C	H	L	P	R	S	T	U
CC51-H120-	12	GJK	H	H	...	...	...	...	...	...
CC51-H150-	15	GJK	H	H	H	H	...	H	...	...
CC51-H180-	18	GJK	H	H	H	H	H	H	...	...
CC51-H200-	20	GJ	H	H	H	H	H	H	H	...
CC51-H220-	22	GJK	H	H	H	H	H	H	H	...
CC51-H240-	24	GJ	H	H	H	H	H	H	H	...
CC51-H270-	27	FGJK	H	H	H	H	H	H	H	...
CC51--300-	30	FGJ	H	H	H	H	H	H	H	J
CC51--330-	33	FGJK	...	...	H	H	H	H	H	J
CC51--360-	36	FGJ	...	...	H	H	H	H	H	J
CC51--390-	39	FGJK	...	...	H	H	H	H	H	J
CC51--430-	43	FGJ	...	...	...	H	H	H	H	J
CC51--470-	47	FGJK	...	...	...	...	...	H	H	J
CC51--510-	51	FGJ	...	...	...	...	...	H	H	J
CC51--560-	56	FGJK	...	...	...	...	...	...	H	J
CC51--620-	62	FGJ	...	...	...	...	...	...	H	J
CC51UJ680-	68	FGJK	...	...	...	...	...	...	...	J
CC51UJ750-	75	FGJ	...	...	...	...	...	...	...	J
CC51UJ820-	82	FGJK	...	...	...	...	...	...	...	J

1/ Complete type designation will include letter symbols to indicate capacitance tolerance, and where applicable, temperature coefficient, and tolerance envelope for temperature coefficient.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20.
For capacitors with values of 20 pF or less, the temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

Life: In accordance with MIL-C-20, operating condition 1.

Marking:
Method I - In accordance with MIL-C-20, except for "JAN" brand.
Method II - In accordance with MIL-C-20 and as shown on figure 3.

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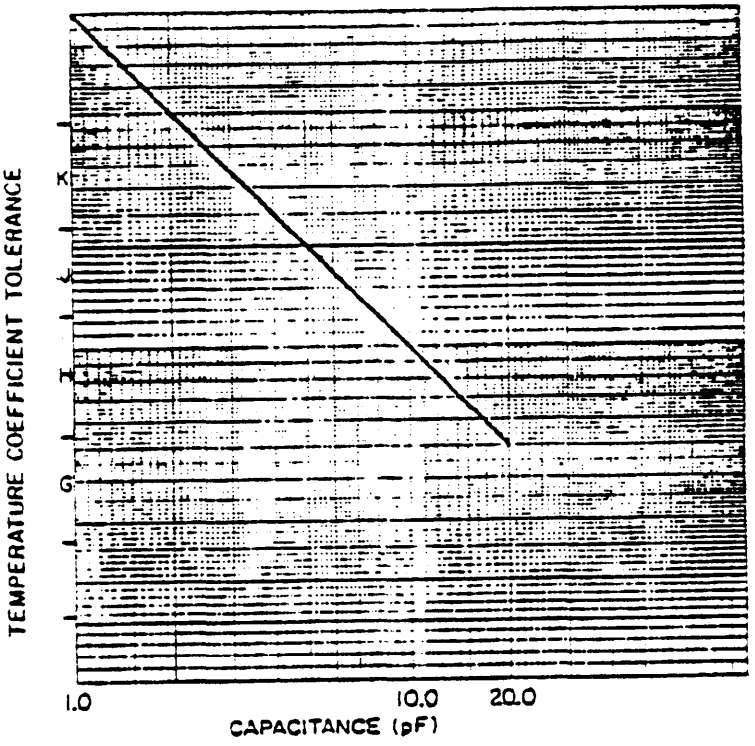
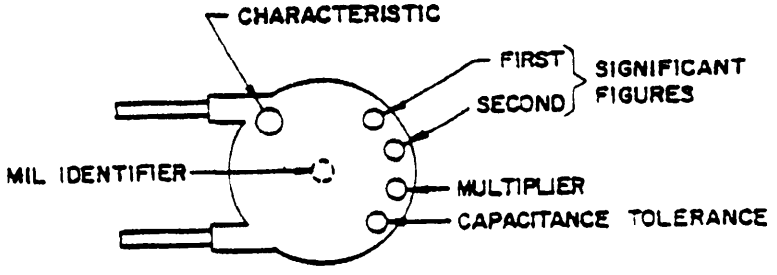


FIGURE 2. Limits for characteristic measurement.



1. The characteristic, capacitance, and capacitance tolerance shall be indicated on the capacitor by five spots of color. The color spots shall be applied so that the designation may be read clockwise. The diameter of the characteristic spot shall be at least twice that of any other spot, or the space between the first and last spots shall be at least three times the space between adjacent spots.

2. The MIL identifier (black spot) shall be applied to the capacitor body approximately in the center and on the reverse side of the capacitor body to which the color code is applied.

FIGURE 3. Color-code marking.

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Navy - EC
Air Force - 11

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Navy - OS
Air Force - 80
DSA - ES

User activities:
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Navy - AS, SH, MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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